

Supplementary Tables 5, 7, and 8

Table 5 (manuscript Table 4), Table 7 (manuscript Table 2), and Table 8 (manuscript Table 5). Bold values match the README layout.

Table 5

Restoration pipeline stage (Table 5)	Hyperparameters	mIoU (%)	mTC (%)
Raw input (no restoration)	—	65.4	77.2
+ Stage 1: flare suppression	$\tau_{\text{flare}} = 220$	67.3	79.5
+ Stage 1: flare suppression	$\tau_{\text{flare}} = 240$	68.6	81.0
+ Stage 1: flare suppression	$\tau_{\text{flare}} = 260$	67.9	79.4
+ Stage 2: DCP dehazing (guided)	$\sigma = 2.0$	70.2	83.2
+ Stage 2: DCP dehazing (guided)	$\sigma = 3.0$	71.4	84.6
+ Stage 2: DCP dehazing (guided)	$\sigma = 4.0$	70.6	83.0
+ Stage 3: CLAHE & gamma	$\gamma = 0.8$	71.2	85.9
+ Stage 3: CLAHE & gamma	$\gamma = 1.4$	73.1	86.8
+ Stage 3: CLAHE & gamma	$\gamma = 2.2$	71.6	86.2
Tri-stage photometric pipeline	—	74.8	88.4

Tri-stage restoration ablation and hyperparameter sweeps (manuscript Table 4).

Table 7

Method (Table 7)	mIoU (%)	mTC (%)	FPS
SAM (per-frame zero-shot)	62.5	64.1	12.3
SAM 2 (streaming video)	71.6	85.1	22.4
Restoration + SAM 3 (causal PCS) — ours	74.8	88.4	28.6

Comparative benchmark excerpt (manuscript Table 2).

Table 8

Framework (Table 8)	Restoration (ms)	Seg. (ms)	FPS	Peak VRAM (MB)
YOLOv11s + SAM	5.8	92.4	10.2	6480
Grounding DINO + SAM	5.8	128.2	7.8	7920
Proposed causal pipeline	5.8	29.1	28.6	4005

Embedded efficiency, Jetson AGX Orin FP16 (manuscript Table 5).